

Effectiveness of Sitagliptin in Indian Patients with T2DM from a Real-World Retrospective EMR Based Study: A Post-Hoc Analysis in Treatment-Naive Patients

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Background

- According to the 2021 ICMR-INDIAB-17 national cross-sectional study, diabetes prevalence in India reached 11.4%, with an additional 15.3% of the population diagnosed with prediabetes,¹ notably Type 2 diabetes mellitus (T2DM) being the most prevalent form, accounting for 90-95% of all diabetes cases.²
- Sitagliptin is a dipeptidyl peptidase 4 (DPP-4) inhibitor, which inhibits the DPP-4 enzyme and deactivates incretin hormones such as glucagon like peptide-1 (GLP -1), resulting in increased insulin secretion in a glucose-dependent manner.³
- Studies have demonstrated reduction in proinsulin:insulin ratio and increase in HOMA-β parameters with use of sitagliptin which suggest improvement in β cell function.⁴
- Sitagliptin is a useful treatment option for patients with T2DM, as it improves glycaemic control with a low incidence of hypoglycaemia and less adverse effects.⁵

Aim

- This real-world, retrospective observational study, using electronic medical records (EMR), was conducted to evaluate the effectiveness of sitagliptin in Indian patients with T2DM.
- This post-hoc analysis represents data of treatment-naive patients who received sitagliptin and had baseline HbA1c between ≥ 7.5% and ≤ 11%.

Methodology

- This real-world, retrospective, observational, EMR based study retrieved anonymised and aggregated data from January 2017 to March 2023 from EMR of patients who were prescribed with sitagliptin alone or with other oral anti-diabetic medications (Figure 1).
- Study was approved by Independent Ethics Committee and registered on CTRI (CTRI no: CTRI/2023/10/058366).

Inclusion criteria

- Male and female patients with T2DM ≥18 years of age.
- Patients prescribed with sitagliptin alone or with other oral antidiabetic medications OR patients prescribed with sitagliptin + metformin FDC alone or with other oral anti-diabetic medications.
- Patients with EMR data captured for more than one visit.

Exclusion criteria

- Patients with type-1 diabetes mellitus.
- Patients on insulin or any other injectable antidiabetic medication like GLP-1 agonists.
- Patients with incomplete medical records with respect to the required study parameters.

Primary Endpoint

- Mean change in HbA1c from baseline to 3 months in patients with HbA1c ≥7% in sitagliptin arm.

Secondary Endpoint

- Mean change in HbA1c from baseline to 3 months in treatment-naive patients with baseline HbA1c ≥7.5% in sitagliptin arm.
- Mean change in PPBG and FBG from baseline to 3 months in treatment-naive patients with baseline HbA1c ≥7% in sitagliptin arm.
- Proportion of patients achieving HbA1c <7% in sitagliptin arm.

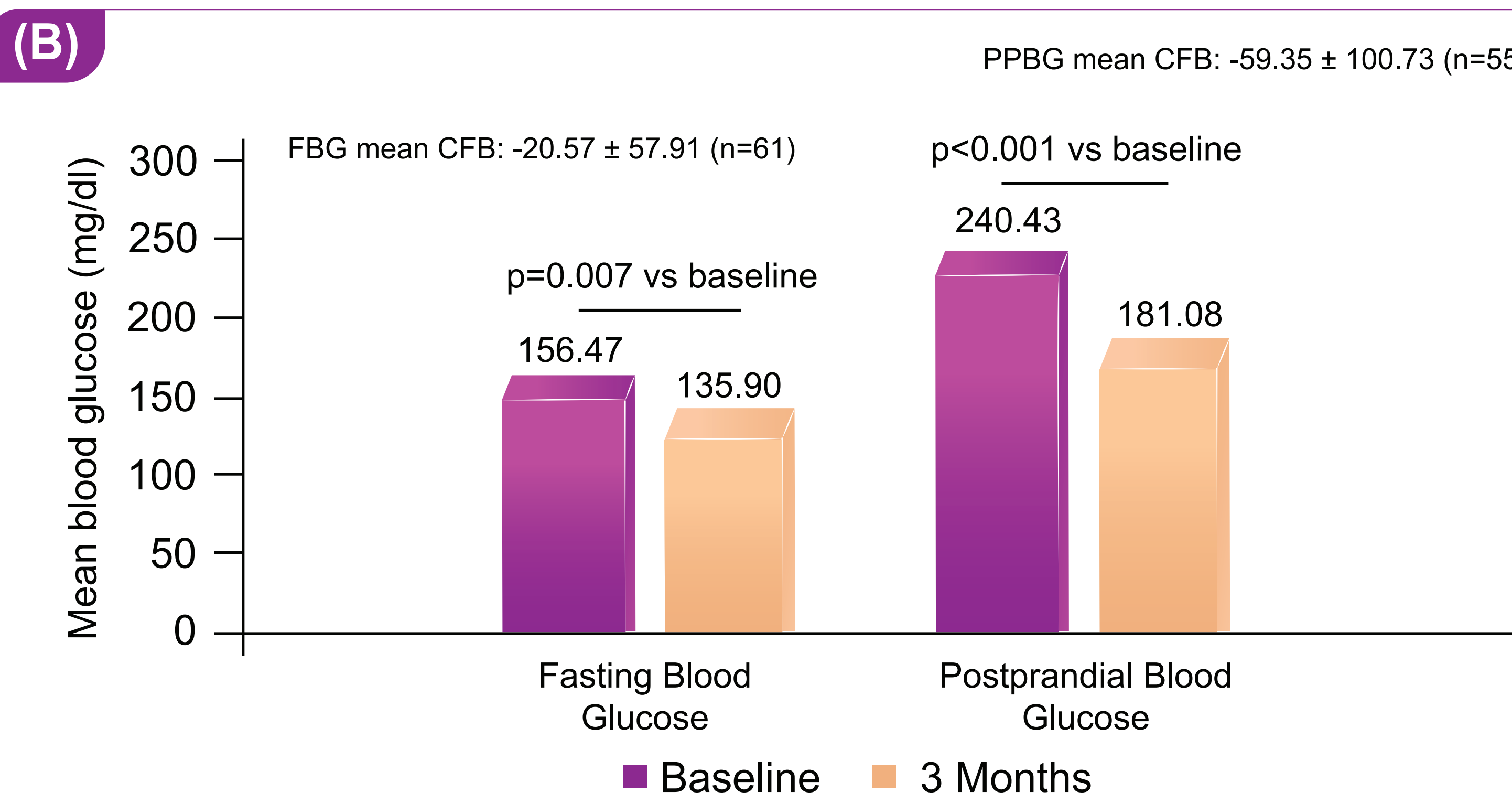
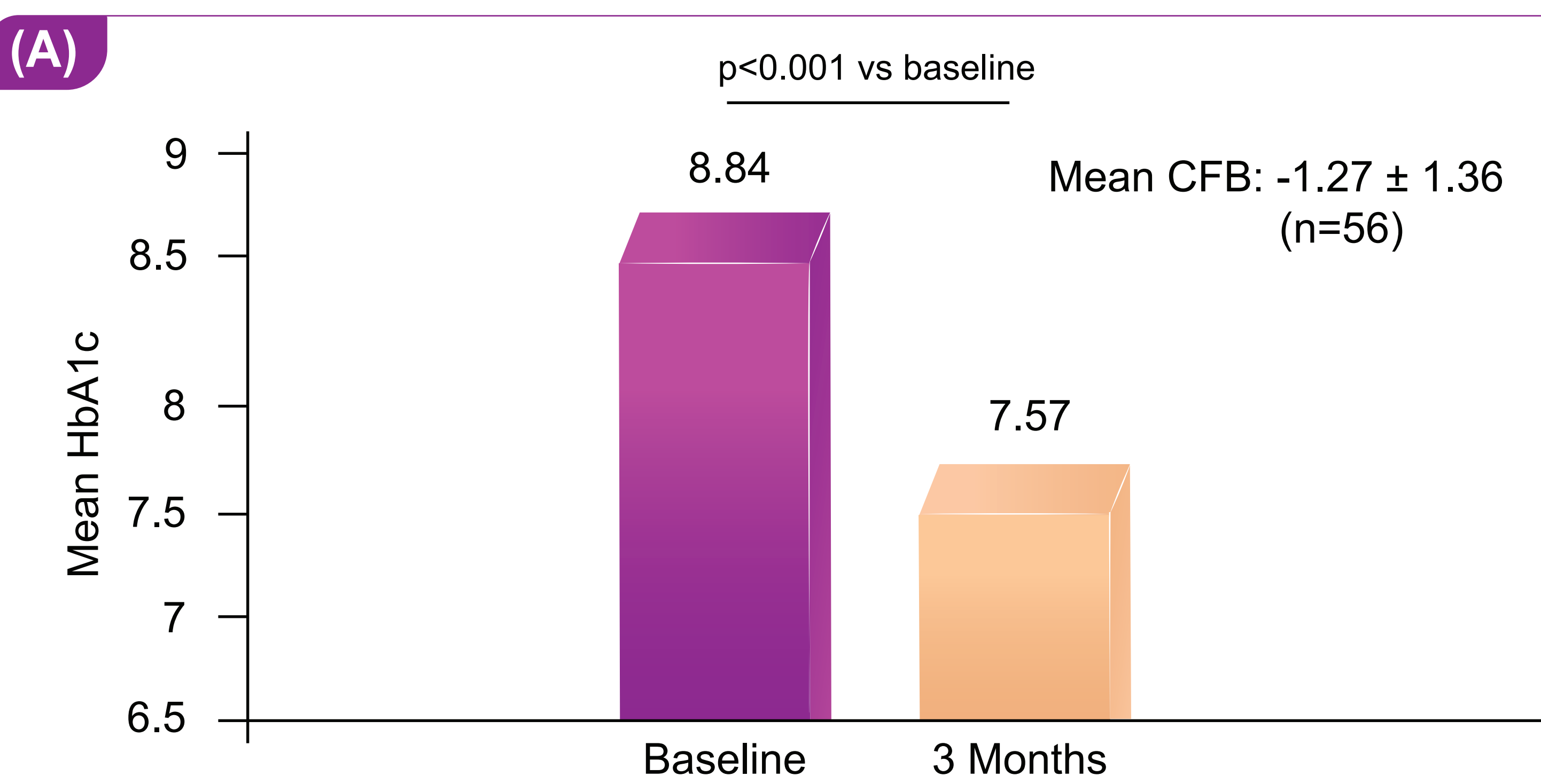
Results

Figure 1: Patient Flow



- 2,794 patients met the eligibility criteria (HbA1c ≥7%; prescribed with sitagliptin in any visit other than the baseline visit on EMR platform and had visits as per the study time points) and 301 of them were treatment naive (Figure 1).

Figure 2: Mean changes in (A) HbA1c, (B) FBG and PPBG from baseline to 3 months



- Mean HbA1c at baseline was 8.84 ± 1.04% (Figure 2).
- A significant reduction in HbA1c levels (-1.27 ± 1.36) was observed over a period of 3 months (p<0.001) (Figure 2).
- Both FBG (156.47 ± 53.35 to 135.90 ± 47.76; CFB: -20.57 ± 57.91, p=0.007) and PPBG (240.43 ± 89.87 to 181.08 ± 66.50; CFB: -59.35 ± 100.73, p<0.001) levels significantly decreased from baseline (Figure 2).
- Proportion of patients achieving HbA1c <7% at 3 months was 32.10 % (n=18/56).

Limitation

- This was a single arm analysis of EMR data captured on digital platform, and the database that we used does not capture safety data.

Conclusion

- This post-hoc analysis demonstrated the effectiveness of sitagliptin in significantly improving HbA1c levels in treatment-naive patients with T2DM who had baseline HbA1c between ≥ 7.5% and ≤ 11.0%.
- Additionally, sitagliptin significantly improved FBG and PPBG in this subset of patients.
- These results emphasize the role of sitagliptin as valuable therapeutic option in managing diabetes in treatment naïve T2D patients.

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